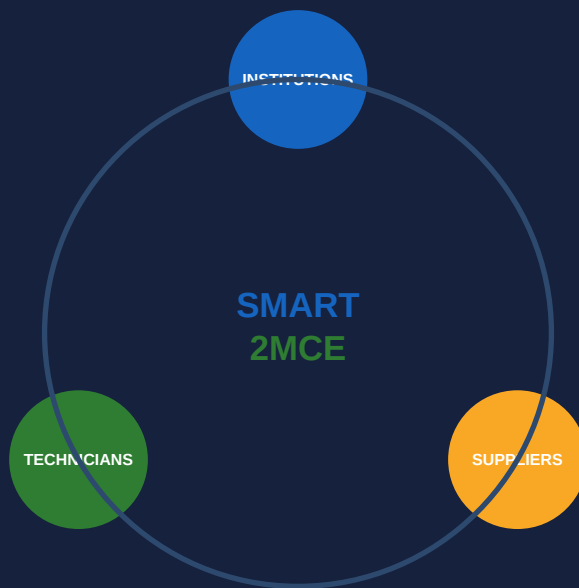




SMART-2MCE

Smart Multi-Maintenance & Multi-Connect Ecosystem

Full Ecosystem Documentation — Version 1.0
Rwanda, 2026



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1. EXECUTIVE SUMMARY & VISION

SMART-2MCE — Smart Multi-Maintenance & Multi-Connect Ecosystem — is a national-level digital platform engineered for Rwanda and designed to scale across Africa. It unifies three economic actors — Institutions, Technicians, and Suppliers — inside one intelligent, structured, and transparent service marketplace. Every maintenance need, device registration, spare-parts procurement, and job opportunity flows through a single, role-aware platform governed by strict workflow rules, real-time monitoring, and AI-powered analytics.

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Mission Statement

SMART-2MCE Mission:

To connect Rwanda's institutions, unemployed graduates, and suppliers into one transparent, intelligent, and accountable maintenance service ecosystem — reducing costs, creating jobs, and improving public and private operational efficiency.

Strategic Goals

**Reduce
Maintenance
Delays**

Smart task deadlines, countdown timers, and late-task auto-alerts ensure no request goes unanswered.

**Create Graduate
Jobs**

The technician marketplace links unemployed graduates directly to real, paid maintenance opportunities.

**Cut Procurement
Costs**

Supplier bidding and competitive quoting prevent overpricing across all sectors.

**Ensure
Accountability**

Full audit trail, ratings, and role-scoped communication create a trusted ecosystem.

2. PROBLEM STATEMENT

The following seven critical challenges exist across Rwanda's public institutions, private companies, NGOs, and civil society — all directly solved by SMART-2MCE.

#	Problem	Impact Description	Affected
01	No Asset Visibility	Devices and materials are unregistered; no one knows what exists, where it is, or its condition.	ALL SECTORS
02	Maintenance Delays	Institutions lack in-house technicians. Repairs wait days or weeks with no tracking or accountability.	PUBLIC & NGO
03	Zero Accountability	No audit trail. Nobody knows who repaired what, when, how, or at what cost.	ALL SECTORS
04	Graduate Unemployment	Thousands of IT, electrical, and mechanical graduates cannot access service opportunities despite available work.	INDIVIDUALS
05	Expensive Procurement	No competitive supplier market. Institutions overpay for spare parts due to lack of price transparency.	ALL SECTORS
06	Broken Communication	Institutions and service providers have no structured channel. WhatsApp and phone calls lead to disputes and loss.	ALL SECTORS
07	Manual, Paper-Based Workflows	Paper requests, handwritten logs, and verbal approvals cause data loss, delays, and impossibility to report.	PUBLIC SECTOR



3. PLATFORM OVERVIEW & IDENTITY

Full Name	Smart Multi-Maintenance & Multi-Connect Ecosystem
Brand Name	SMART-2MCE
Version	1.0 (April 2026)
Platform Type	Multi-sided Digital Service Marketplace + Asset Management System
Technology	React.js + Node.js (Express) + PostgreSQL + Socket.IO
Primary Colors	Blue (Trust & Technology) Green (Growth & Maintenance) Yellow (Energy & Suppliers)
Target Market	Rwanda — scalable to East Africa and beyond
Design Philosophy	Service-oriented (like Irembo.rw) — users request services, not just use a system
Key Tagline	"Connecting Maintenance. Empowering Ecosystems."

System Modules

Organization Engine	Manages all institution types and their devices/materials
Service Request Engine	Full lifecycle from request creation to closure
Technician Marketplace	Graduate job discovery, proposals, and negotiation
Supplier Marketplace	Competitive spare-parts bidding and procurement
Time & Alert Engine	Countdown timers, deadline tracking, late-task alerts
Communication Engine	Task-scoped, role-restricted messaging
AI Analytics Module	Charts, performance insights, predictive alerts
Learning Center	Training content for all users





4. SYSTEM ARCHITECTURE

SMART-2MCE is built on a modern three-tier architecture with a React.js frontend, Node.js + Express backend API, and a PostgreSQL relational database. Real-time features are powered by Socket.IO. The system is multi-tenant: each organization has isolated data within a shared platform infrastructure.



★ Multi-Tenant | Role-Based Access Control | Real-Time via Socket.IO | JWT Authentication

Data Flow Summary

Actor	Sends Request To	Response From	Real-Time?
Institution Admin	Node.js API (REST)	React UI update	Yes — task status, notifications
External Technician	Node.js API (REST)	Marketplace data	Yes — new task alerts
Supplier	Node.js API (REST)	Quote confirmations	Yes — order updates
System (Scheduler)	PostgreSQL query	Notification push	Yes — countdown & late alerts



5. USER ROLES & ACTORS

SMART-2MCE has five distinct actor types. Every actor sees a role-specific dashboard and can only access features, data, and communication channels that belong to their role.

INSTITUTION ADMIN

Public hospitals, government offices, universities, NGOs, private companies — any registered organization administrator.

- Register and manage organization profile and location
- Register all devices (with ID e.g. KRH/MH/001) and materials (by room)
- Create service requests with deadline, priority, and required skills
- Assign tasks internally or publish to the external marketplace
- Review technician proposals and supplier quotes
- Monitor countdown timers and task progress
- Verify completed work and issue ratings
- Access full analytics dashboard

INTERNAL TECHNICIAN

Full-time maintenance staff employed directly by an institution. First priority for all tasks.

- Receive directly assigned tasks from their institution admin
- View linked device ID/room and full problem description
- Update task progress: Started → In Progress → Completed
- Raise spare parts requests when materials are needed
- Build internal performance record and rating history

EXTERNAL TECHNICIAN (Graduate)

Independent professionals and unemployed graduates seeking maintenance job opportunities through the marketplace.

- Create a public profile with skills, location, and certifications
- Browse available tasks filtered by location and skill type
- Submit proposals: proposed price, timeline, and cover message
- Negotiate terms through task-scoped communication
- Receive formal acceptance before starting any work
- Update progress in real-time during task execution
- Build public reputation through ratings and job history



SUPPLIER

Businesses and vendors providing spare parts, materials, and equipment to institutions.

- Register with product categories and delivery coverage area
- Browse open spare-part procurement requests
- Submit competitive quotes: price, delivery time, notes
- Fulfill confirmed orders and update delivery status
- Build reputation through ratings and performance metrics

PLATFORM AI ENGINE

The intelligent backend that orchestrates all workflows, enforces rules, and drives intelligence.

- Enforce task lifecycle rules and status transitions automatically
- Run countdown timers and detect late tasks in real-time
- Send multi-level notifications and escalation alerts
- Match tasks to suitable technicians by skill and proximity
- Generate analytics charts and performance reports
- Power the AI assistant and learning center recommendations

6. LANDING PAGE & PUBLIC ONBOARDING

The landing page is the first point of contact for all visitors. It communicates the platform's value proposition, showcases the three actor types, and guides users to register or log in. The design follows a modern, professional aesthetic inspired by platforms like Irembo.rw — clean, structured, and action-oriented.

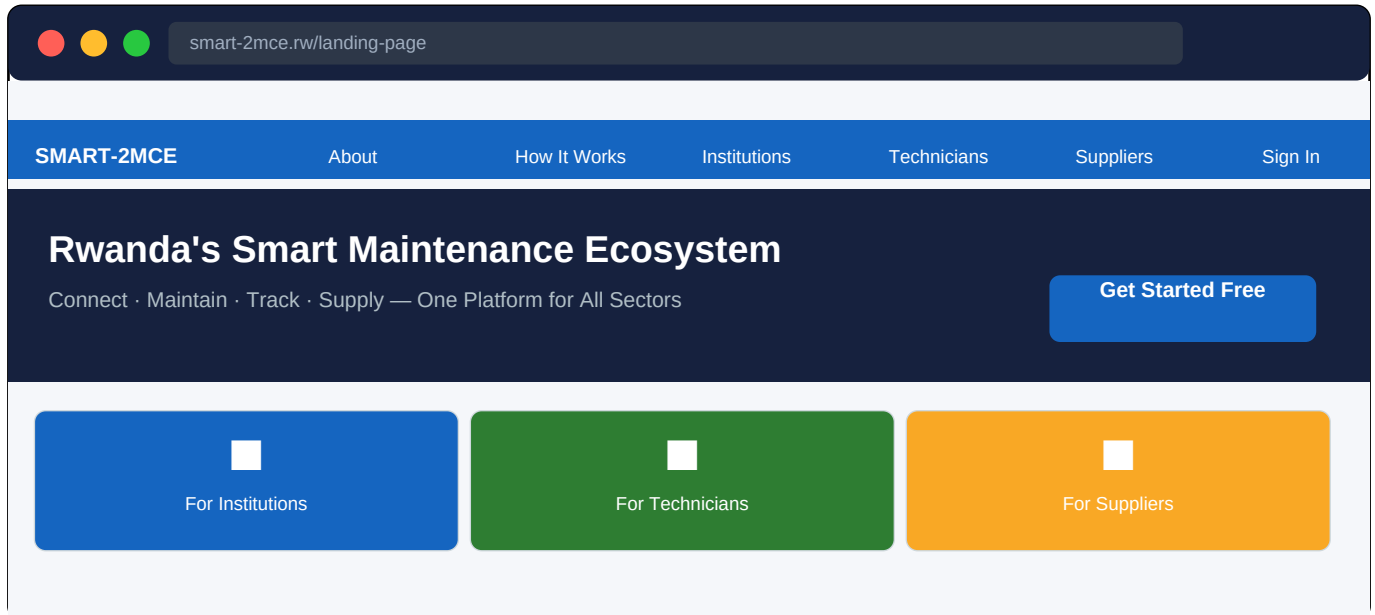


Fig. 1 — SMART-2MCE Public Landing Page

Landing Page Sections

Section	Description
Hero Section	Full-width banner with headline, tagline, and dual CTAs: "Register as Institution" and "Find Work as Technician"
How It Works	3-step visual flow: Post a Need → Find the Right Expert → Track & Complete. Explained with icons and clean prose
Actor Cards	Three feature panels: Institution (blue), Technician (green), Supplier (yellow) — each with entry CTA
Featured Institutions	Real institution names: University of Rwanda, King Faisal Hospital, REB, RURA, MININFRA, BK, MTN Rwanda, Equity Bank Rwanda
Impact Statistics	Live counters: Active Institutions, Tasks Completed, Technicians Registered, Sectors Served
Testimonials	Short quotes from institution admins, technicians, and suppliers about platform value
Footer	Links: About, Contact, Privacy Policy, Terms. Rwanda flag colors. Copyright SMART-2MCE 2026

7. AUTHENTICATION — REGISTER & LOGIN

7.1 Registration Flow

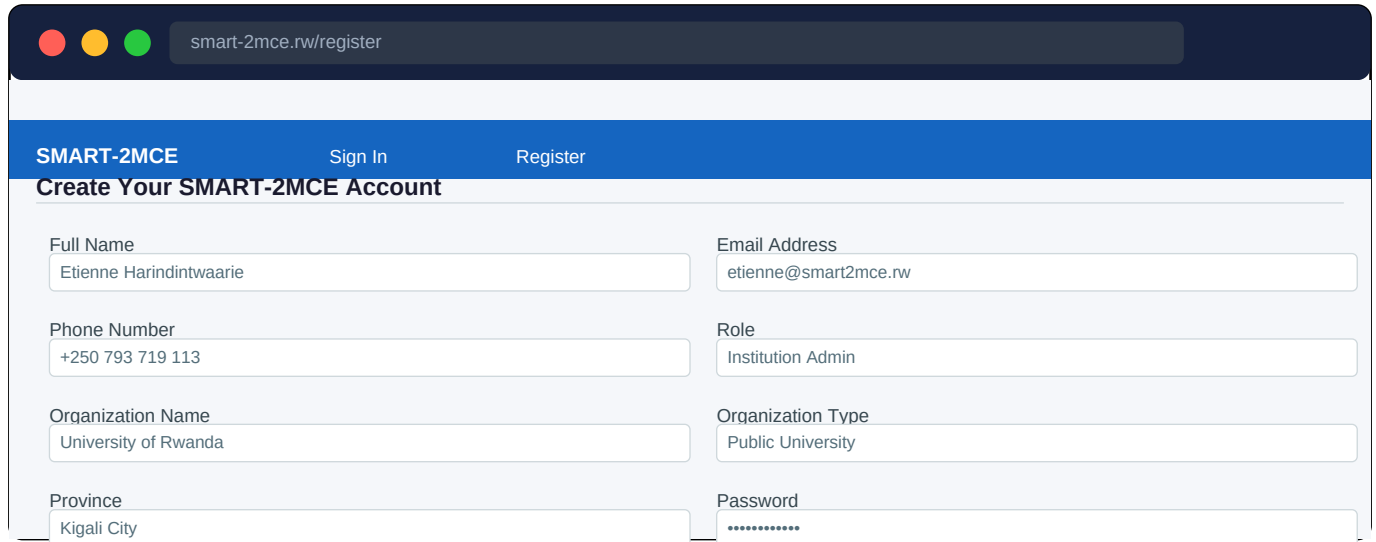


Fig. 2 — Multi-step Registration Form

Registration is multi-step and role-adaptive. The form changes based on the selected role (Institution Admin, External Technician, or Supplier). The location hierarchy (Province → District → Sector → Cell → Village) uses dynamic dependent dropdowns.

Step	Stage	Details
Step 1	Personal Details	Full name, email, phone, password, confirm password
Step 2	Select Role	Institution Admin / External Technician / Supplier — form adapts
Step 3	Organization/Profile	Organization name, type, location hierarchy, OR technician skills/certs
Step 4	OTP Verification	One-time password sent to email or phone — confirms identity
Step 5	Account Active	Profile complete; redirected to role-specific onboarding tour

7.2 Login & Security



smart-2mce.rw/login

SMART-2MCERegisterHelp

Sign In to SMART-2MCE

Email Address
admin@ur.ac.rw

Password

☐ Remember me

Forgot password?

[Sign In]

[Sign in with Google]

OTP: Enter 6-digit code sent to your phone/email to verify identity

Fig. 3 — Secure Login with OTP Two-Factor Authentication

Security Feature	Implementation	Purpose
JWT Tokens	JSON Web Tokens with expiry	Stateless, secure session management
OTP Verification	Time-limited 6-digit code via email/SMS	Two-factor authentication
BCrypt Passwords	Salted hash — never stored in plain text	Password security
Role Middleware	Server-side role check on every API call	Prevent privilege escalation
CSRF Protection	Token-based cross-site request forgery prevention	Encryption
Rate Limiting	Max login attempts per IP per hour	Brute-force prevention



8. INSTITUTION ADMIN DASHBOARD

After login, the Institution Admin sees a professional, information-dense dashboard giving full visibility over all operations. The sidebar navigation is role-specific and the home overview surfaces the most critical data immediately.

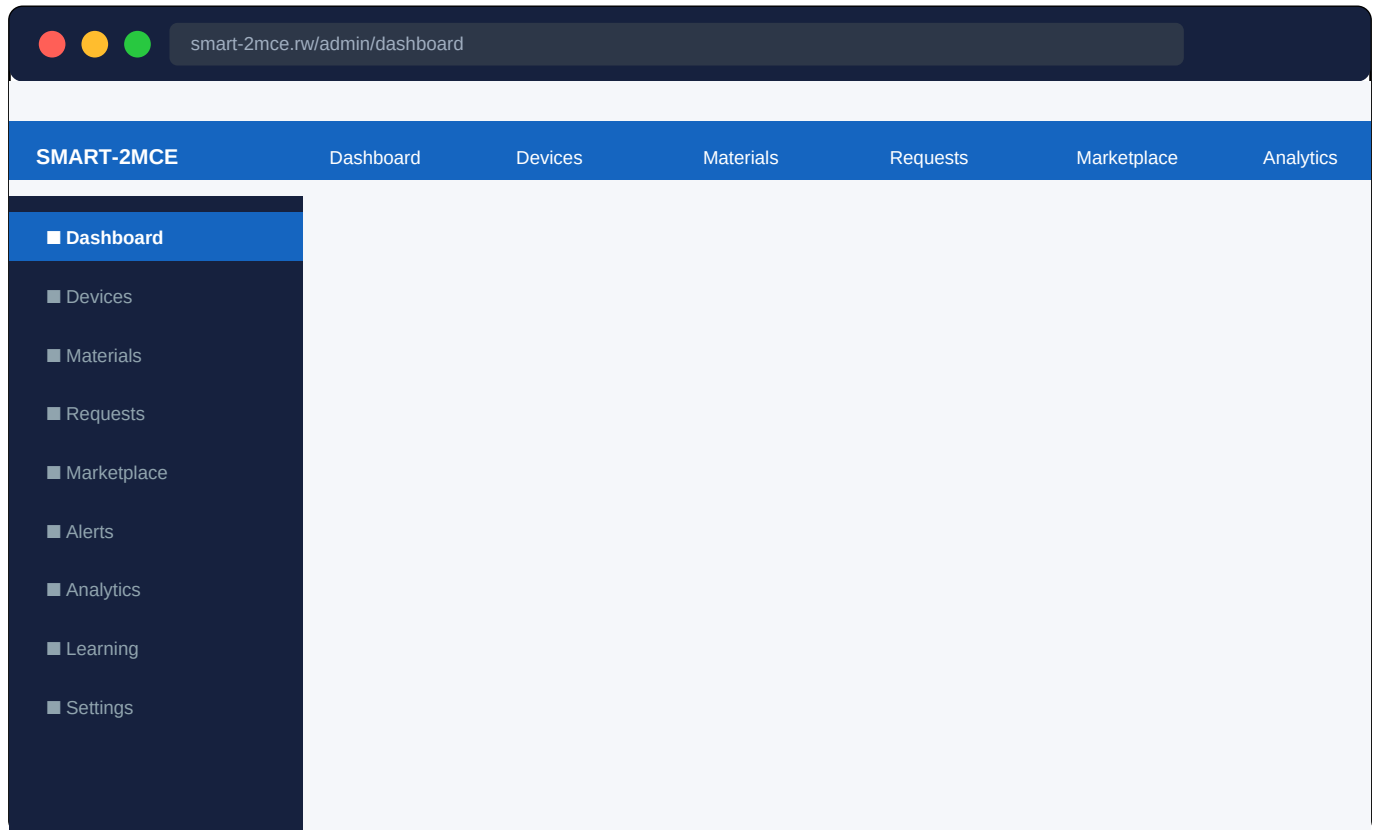


Fig. 4 — Institution Admin Dashboard — Overview

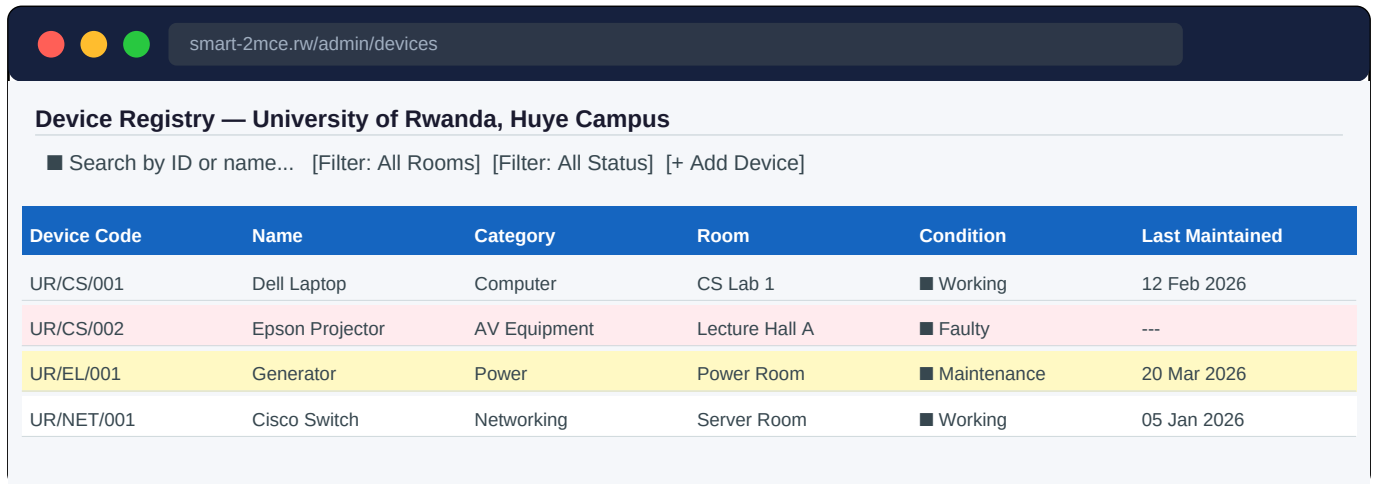
Dashboard Navigation Sections

Section	Contents & Functionality
Overview / Home	KPI cards, recent requests, live countdown timers, late-task alerts, pending approvals
Devices	Full registry — search by ID, filter by room/status/category, add/edit/decommission
Materials	Room-based inventory, quantity tracking, condition updates, damage reports
Service Requests	All requests with lifecycle status, countdown, priority badge, technician assignment
Marketplace	Published announcements, incoming proposals with comparison view, supplier quotes
Notifications	Alert center — read/unread, filter by type, acknowledge and archive
Analytics	Charts: tasks by status, cost trends, technician performance, late-task heatmap
Learning Center	Training modules, guides, device manuals for all staff
Settings	Organization profile, user management, role permissions, notification preferences

9. DEVICE & MATERIAL MANAGEMENT

9.1 Device Registry

Every physical device owned by an institution is registered in SMART-2MCE. Public institutions use structured asset codes (e.g., KRH/MH/001 = Kigali Regional Hospital / Maintenance Dept / Unit 001). Private and NGO organizations may use flexible naming. All devices are searchable, filterable, and link directly to maintenance history.



Device Code	Name	Category	Room	Condition	Last Maintained
UR/CS/001	Dell Laptop	Computer	CS Lab 1	■ Working	12 Feb 2026
UR/CS/002	Epson Projector	AV Equipment	Lecture Hall A	■ Faulty	---
UR/EL/001	Generator	Power	Power Room	■ Maintenance	20 Mar 2026
UR/NET/001	Cisco Switch	Networking	Server Room	■ Working	05 Jan 2026

Fig. 5 — Device Registry with ID-based tracking

9.2 Material Registry (Room-Based)

Materials are tracked by room name rather than unique IDs. This covers furniture, tools, cables, consumables, and semi-permanent items. Quantity and condition are updated whenever a maintenance or damage report is filed.

Field	Description	Example
Material Name	Full name of the item	Network Cable Cat6
Room Name	Room or area location	Server Room B
Quantity	Total units	24 units
Condition	Working / Damaged / Needs Replacement	Needs Replacement (8 units)
Organization	Owning institution	Kigali Independent University
Last Updated	Date of last status change	22 April 2026

9.3 Supported Institutions & Their Asset Systems

The following represent example real-world institutions and how they use the device/material registry:

Institution	Device Code Format	Common Devices Tracked
University of Rwanda (UR)	UR/DEPT/###	IT labs, projectors, generators, networking gear
King Faisal Hospital Kigali	KFH/MED/###	Medical equipment, ICU devices, imaging machines

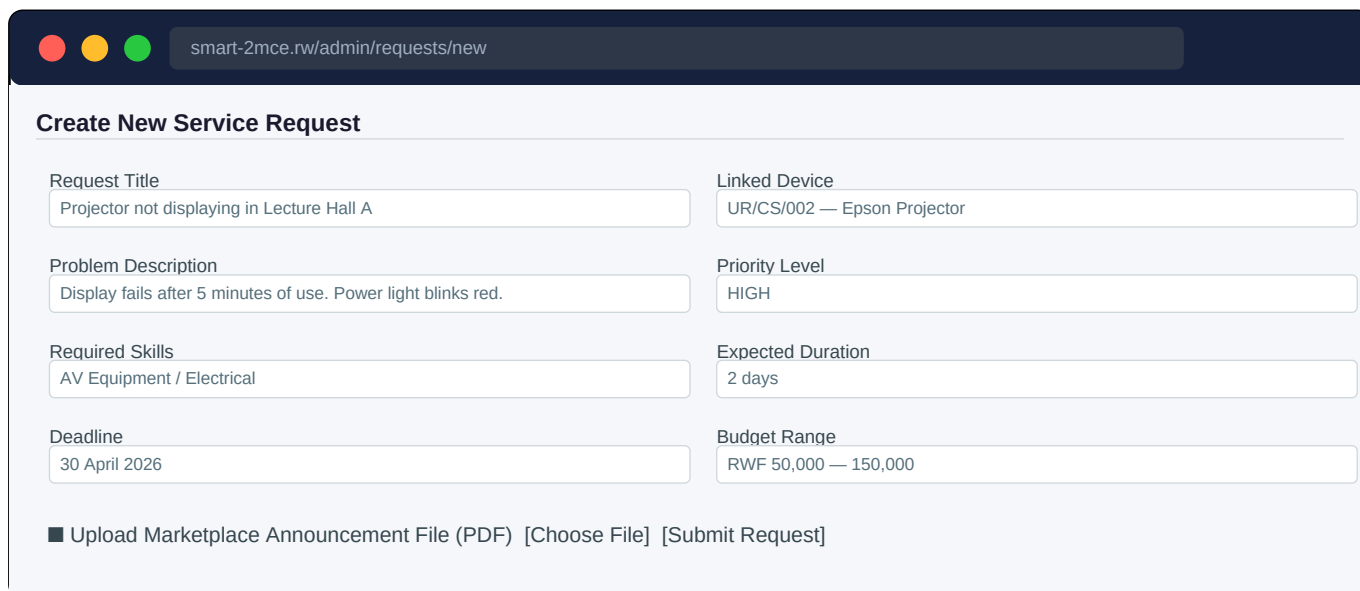


Institution	Device Code Format	Common Devices Tracked
Rwanda Education Board (REB)	REB/OFF/###	Office computers, printers, AV systems
Rwanda Utilities Regulatory Authority	RURA/###/###	Office equipment, monitoring stations
Ministry of Infrastructure (MININFRA)	MINF/###/###	Vehicles, machinery, office systems
Bank of Kigali (BK)	BK/BR/###	ATMs, computers, security systems
MTN Rwanda	MTN/###/###	Network equipment, office devices
Equity Bank Rwanda	EBR/BR/###	Banking terminals, servers, printers
Kigali Independent University (KIU)	KIU/###/###	Computers, lab equipment, library systems
Rwanda Polytechnic (RP)	RP/###/###	Workshop tools, lab equipment



10. SERVICE REQUEST WORKFLOW

The service request is the core unit of the entire SMART-2MCE ecosystem. It represents a single maintenance need, links a device or material, triggers all marketplace activity, and drives all notifications, assignments, and ratings. The lifecycle has 11 stages.



Create New Service Request

Request Title: Projector not displaying in Lecture Hall A

Linked Device: UR/CS/002 — Epson Projector

Problem Description: Display fails after 5 minutes of use. Power light blinks red.

Priority Level: HIGH

Required Skills: AV Equipment / Electrical

Expected Duration: 2 days

Deadline: 30 April 2026

Budget Range: RWF 50,000 — 150,000

■ Upload Marketplace Announcement File (PDF) [Choose File] [Submit Request]

Fig. 6 — Service Request Creation Form

10.1 Full Task Lifecycle (11 Stages)

1	CREATED	Institution admin submits the service request with all required fields
2	INTERNAL REVIEW	System checks if internal technicians exist. Admin may assign directly
3	OPEN — MARKETPLACE	If no internal tech available, task appears in external marketplace
4	UNDER APPLICATION	One or more external technicians have submitted proposals
5	NEGOTIATION	Institution and selected technician negotiate price, timeline, scope
6	ASSIGNED	Institution formally accepts a technician — task is locked
7	IN PROGRESS	Technician actively performing maintenance — progress updated
8	WAITING PARTS	Work paused — spare parts requested from supplier marketplace
9	COMPLETED	Technician marks work done — awaiting institution verification
10	VERIFIED	Institution confirms quality and accepts the completed work
11	CLOSED	Task archived — both parties rate each other — history saved
■	LATE (Auto-Flag)	Automatic flag when deadline passes without COMPLETED status



10.2 Priority System

Priority	Trigger	Response Expectation	Alert Behavior
LOW	Minor issue, no operational impact	Within 7 days	Single reminder at 80% time
MEDIUM	Partial disruption to operations	Within 3 days	Reminders at 50% and 80%
HIGH	Significant disruption	Within 24 hours	Immediate + hourly reminders near deadline
CRITICAL	Operations fully halted	Within 4 hours	Immediate admin escalation alert

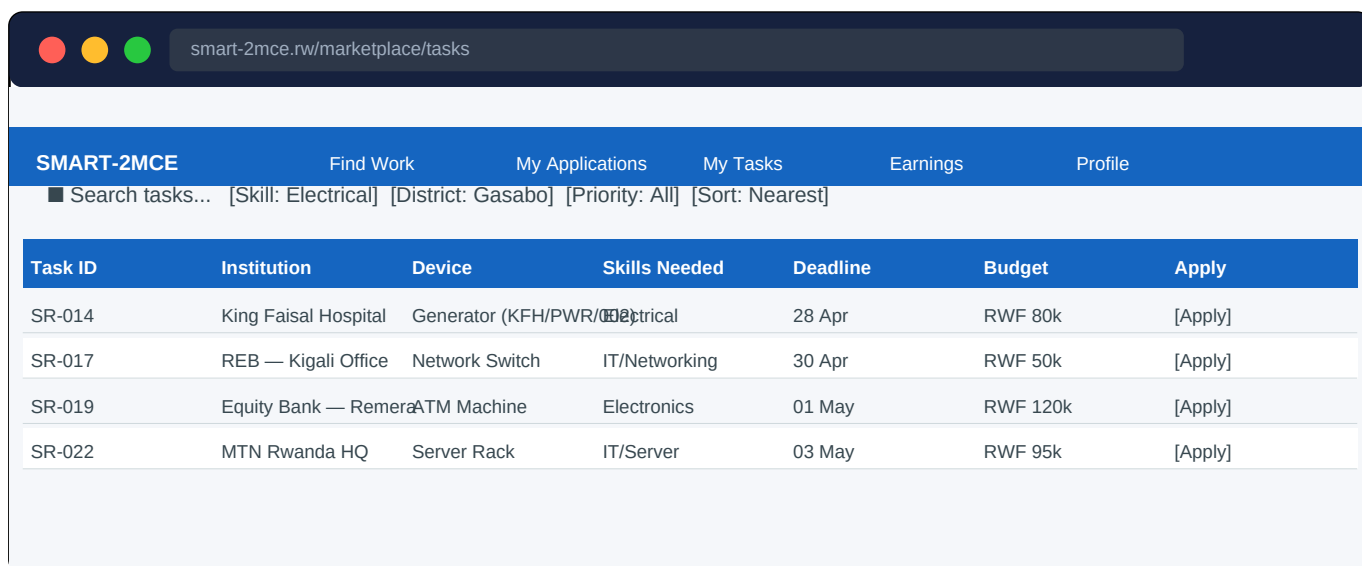


11. TECHNICIAN MARKETPLACE

The Technician Marketplace is the employment innovation at the heart of SMART-2MCE. It solves graduate unemployment by connecting skilled individuals to real maintenance opportunities — transparently, competitively, and on their own terms. No task is auto-assigned. Every technician must apply, and the institution makes the final choice.

Core Rule — No Auto-Assignment:

External technicians browse available tasks and submit proposals voluntarily. The system NEVER automatically assigns work. This preserves technician choice, institution control, and marketplace integrity.



Task ID	Institution	Device	Skills Needed	Deadline	Budget	Apply
SR-014	King Faisal Hospital	Generator (KFH/PWR/100kVA)	Electrical	28 Apr	RWF 80k	[Apply]
SR-017	REB — Kigali Office	Network Switch	IT/Networking	30 Apr	RWF 50k	[Apply]
SR-019	Equity Bank — Remera	ATM Machine	Electronics	01 May	RWF 120k	[Apply]
SR-022	MTN Rwanda HQ	Server Rack	IT/Server	03 May	RWF 95k	[Apply]

Fig. 7 — Technician Marketplace — Browse Available Tasks

11.1 Proposal Submission



Apply for Task SR-014 — Generator Repair, King Faisal Hospital

Your Proposed Price (RWF): 75,000

Estimated Duration: 6 hours

Availability Start Date: 27 April 2026

Relevant Skills: Electrical Engineering, Generator Maintenance, 4 years exp.

Cover Message: "I have completed 3 generator repairs for hospitals in Kigali this year..."

[Attach Certificate / Portfolio] [Submit Proposal]

Fig. 8 — Technician Proposal Submission

11.2 Negotiation Flow

Step	Actor	Action
1	External Technician	Submits proposal with price, time, and cover message
2	Institution Admin	Reviews all proposals side-by-side — ratings, price, time
3	Institution Admin	Opens negotiation thread with preferred candidate
4	Technician	Responds in task-scoped chat — adjusts terms if needed
5	Institution Admin	Issues formal acceptance — task status changes to ASSIGNED
6	Other Applicants	Receive automatic rejection notification
7	System	Locks task to accepted technician — countdown begins

11.3 Technician Profile & Reputation

Profile Element	Description
Skills	List of technical skills: IT, Electrical, Mechanical, Plumbing, AV, Medical Equipment, etc.
Certifications	Uploaded qualifications and diplomas (verified by platform)
Location	Province, District, Sector — used for proximity matching and job filtering
Rating	Weighted average of all institution ratings (1–5 stars)
Completed Tasks	Total jobs done, broken down by category and sector
Response Rate	Percentage of proposals that received responses within 24 hours
Late Task Rate	Percentage of assignments completed past deadline — affects rating

12. SUPPLIER MARKETPLACE

The Supplier Marketplace enables institutions to source spare parts and materials competitively. Suppliers register their catalog, respond to open procurement requests, and fulfill confirmed orders. Competitive quoting prevents overpricing and gives institutions full control over selection.

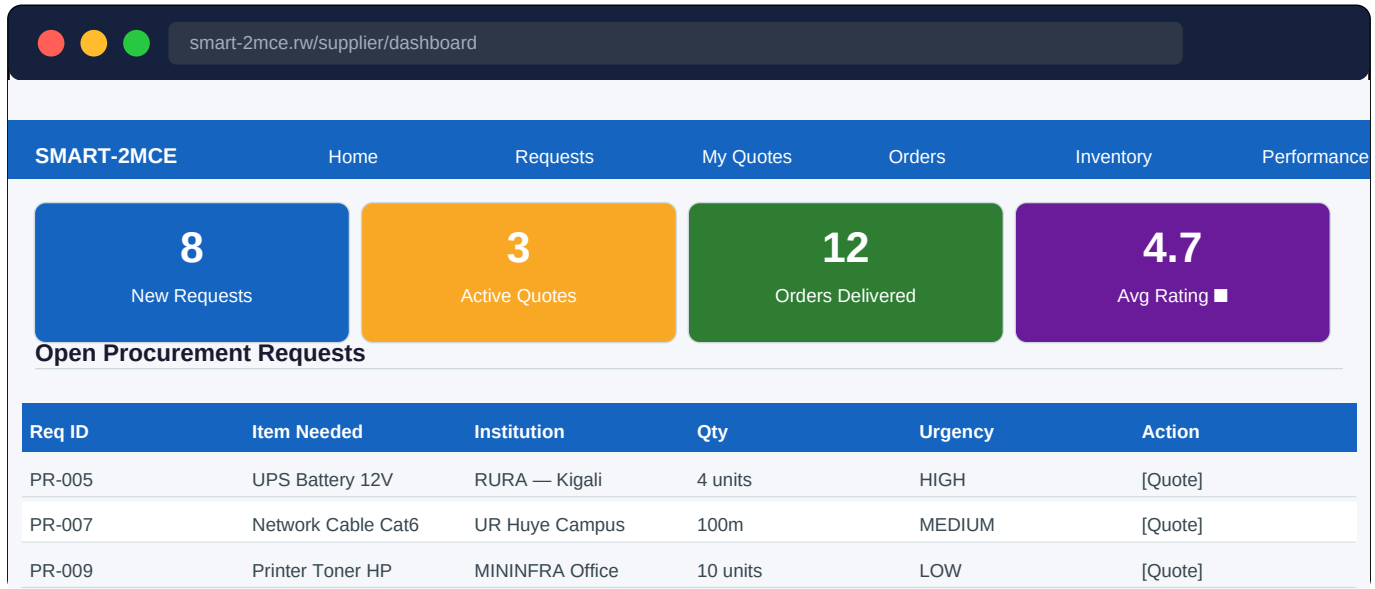


Fig. 9 — Supplier Dashboard

Supplier Quote & Order Flow

Stage	Action	By Whom
Request Posted	Institution raises spare-part request linked to an active task	Institution Admin
Suppliers Notified	System alerts all suppliers matching the item category and location	Platform
Quote Submitted	Supplier enters price, delivery time, brand/model, and quantity	Supplier
Comparison View	Institution sees all quotes ranked by price, delivery time, and rating	System
Quote Accepted	Institution selects best supplier — order confirmed automatically	Institution Admin
Fulfillment	Supplier prepares order and updates status: Prepare → In Transit → Delivered	Supplier
Receipt Confirmed	Institution confirms delivery — payment reference logged	Institution Admin
Rating	Institution rates supplier on quality, speed, and accuracy	Institution Admin

13. SMART TIME & NOTIFICATION ENGINE

Every task in SMART-2MCE is time-bound. The time engine monitors all active tasks continuously, triggering alerts, escalations, and status changes automatically. This is the operational backbone that prevents delays and holds everyone accountable.

13.1 Time Fields Per Task

Field	Type	Description
created_at	Timestamp	Auto-set when request is submitted
expected_duration	Integer	Admin-set: estimated hours or days to resolve
deadline	Timestamp	Final completion target — shown as live countdown to all parties
countdown_timer	Computed	deadline - current_time — refreshed every minute via Socket.IO
actual_completion	Timestamp	Set when technician marks COMPLETED
late_flag	Boolean	Auto-set TRUE when current_time > deadline AND status != COMPLETED
delay_minutes	Integer	actual_completion - deadline in minutes — used in analytics

13.2 Multi-Level Alert Schedule

Alert Level	Trigger Condition	Who Receives	Notification Type
Reminder	50% of time elapsed — task not yet started	Technician	In-app info
Warning	80% of time elapsed — task in progress	Technician + Admin	In-app + Email
Pre-LATE	95% of time — task not COMPLETED	Technician + Admin	Push + Email
LATE	Deadline passed, status != COMPLETED	Technician + Admin	Critical push + Email + badge
Escalation	LATE for 24+ hours — no response	Admin + Platform team	Urgent email escalation
New Task	Task published to marketplace	Matching technicians	In-app + Email digest
Proposal	Technician submits proposal	Institution Admin	In-app
Accepted	Institution accepts proposal	Technician	In-app + Email
Parts Needed	Technician flags spare parts required	Admin + Suppliers	In-app
Delivered	Supplier confirms delivery	Technician + Admin	In-app
Completed	Technician marks task done	Institution Admin	In-app + Email

14. COMMUNICATION ENGINE

SMART-2MCE does not allow free, open messaging. All communication is task-scoped, role-restricted, and permanently recorded. This ensures every conversation is accountable, contextual, and retrievable — similar to how enterprise service platforms like Jira Service Management or Irembo handle inter-party communication.

Channel	Parties	Scope	Purpose
Negotiation Thread	Institution ↔ External Technician	Per task only	Price, timeline, and scope discussion before assignment
Supplier Thread	Institution ↔ Supplier	Per order only	Quote clarification, delivery coordination
Internal Work Channel	Institution Admin ↔ Internal Technician	Per task only	Direct assignment instructions and progress updates
Progress Updates	Technician → Institution	Per task	Real-time work updates: started, blocked, completed
System Notifications	Platform → All Users	Global	Alerts, reminders, status changes — one-way only

smart-2mce.rw/requests/sr-014/negotiate

Negotiation — SR-014 — King Faisal Hospital / Generator Repair

Hi, I am available 27 Apr. My quote is RWF 75,000 for 6 hrs.

Thank you. We would like completion within 4 hrs. Can you adjust?

Yes, I can do 4 hours for RWF 85,000 including parts check.

Agreed. We will issue formal acceptance now.

[Accept & Assign Technician] [Reject Proposal] [Continue Negotiating]

Fig. 10 — Task-Scoped Negotiation Thread

15. EXTERNAL TECHNICIAN DASHBOARD

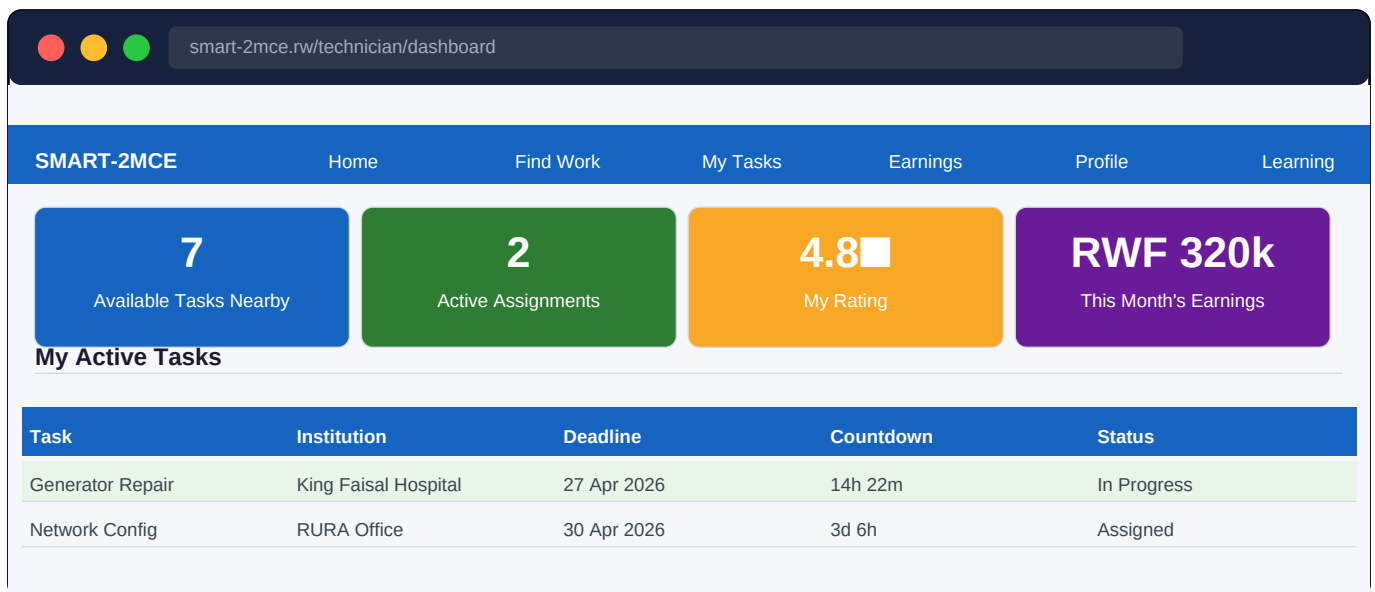


Fig. 11 — External Technician Dashboard

Technician Dashboard Sections

Section	Contents
Home	Available tasks nearby, active assignments, rating, earnings summary, alerts
Find Work	Marketplace browser — filter by location, skill type, urgency, budget
My Applications	Proposals submitted, negotiation status, accepted/rejected history
My Tasks	Current active assignments — countdown timer, progress stages, parts requests
Task History	All completed tasks — ratings received, work log, institution feedback
Profile	Skills, certifications, location, public rating, portfolio
Earnings	Income tracking per task, monthly summary, payment history
Learning	Access to training modules and maintenance guides

17. AI ASSISTANT & ANALYTICS MODULE

SMART-2MCE integrates an AI assistant and a real-time analytics dashboard across all user types. The AI assistant answers questions, guides users through workflows, and provides intelligent suggestions. Analytics charts empower data-driven decisions at every level — institution, technician, supplier, and platform-wide.

17.1 AI Assistant

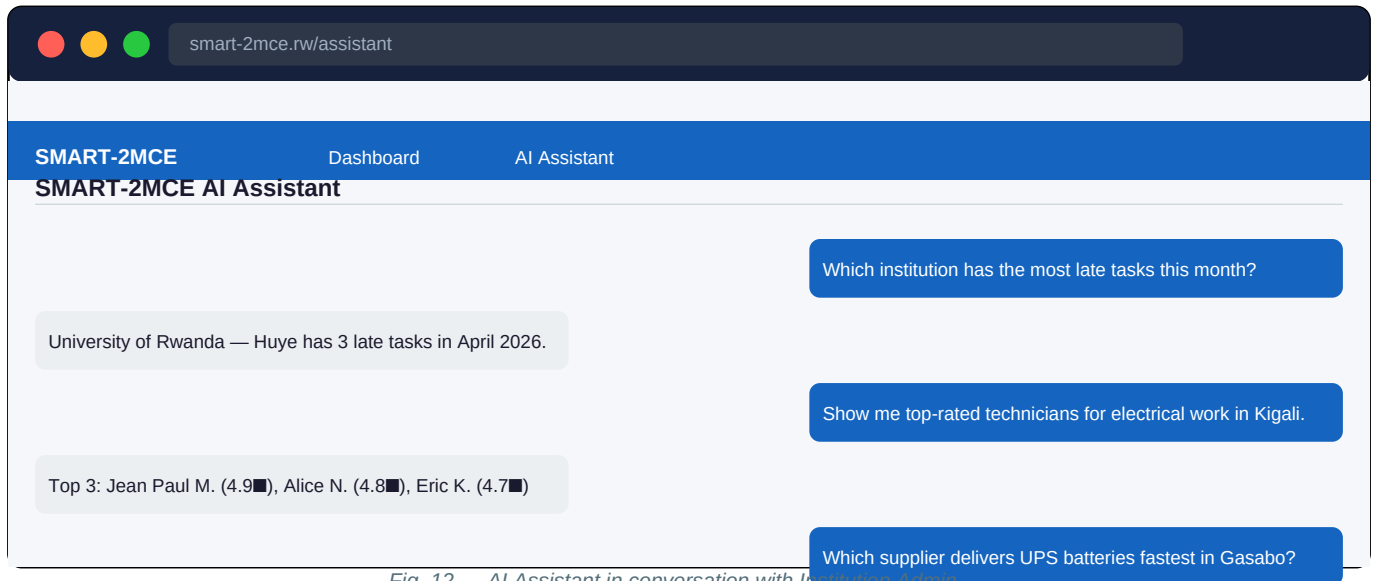


Fig. 12 — AI Assistant in conversation with Institution Admin

AI Assistant Capabilities

Capability	Description
Workflow Guidance	Guides users step-by-step through request creation, marketplace use, and negotiation
Smart Recommendations	Suggests best-matched technicians based on task type, location, and past rating
Supplier Insights	Recommends suppliers by item, delivery speed, price history, and proximity
Performance Queries	Answers natural-language questions about task stats, late rates, costs
Predictive Alerts	Warns: "Device UR/CS/002 has failed 3 times this year — consider replacement"
Institution Suggestions	Lists proposed institutions: UR, KFH, REB, RURA, MININFRA, BK, MTN, Equity Bank, KIU, RP
Learning Guidance	Recommends training modules based on user role and recent activity

17.2 Analytics Dashboard

The analytics module provides real-time visual charts for administrators. All data is live, refreshed via API calls on page load and Socket.IO pushes on state changes.



Chart Type	Data Shown
Tasks by Status (Donut)	Shows distribution: Open, In Progress, Late, Completed, Closed
Monthly Task Volume (Bar)	Month-by-month request count — identifies seasonal maintenance peaks
Cost Trend Line (Line)	Monthly spend on maintenance and procurement — tracks budget health
Technician Performance (Bar)	Completed vs late tasks per technician — identifies top performers
Supplier Delivery Time (Box)	Avg delivery time per supplier — drives future selection
Late-Task Heatmap	District-level heatmap showing where maintenance delays are highest
Device Failure Frequency	Top 10 most-repaired devices — flags candidates for replacement





18. LEARNING CENTER

The Learning Center is a preventive maintenance innovation. By educating all users — staff, technicians, and cleaners — on proper device usage and early fault detection, the platform reduces damage before it happens. This directly saves institutions money and reduces the volume of service requests over time.

Module	Audience	Content
Device Usage Best Practices	All Staff	How to operate and care for computers, projectors, printers, and other devices
Early Fault Detection	All Staff	Signs that a device needs maintenance: sounds, lights, performance indicators
Maintenance Reporting Guide	All Users	How to create a complete, useful service request with photos and descriptions
Technician Skills Modules	Technicians	IT troubleshooting, electrical safety, generator maintenance, network configuration
Procurement Best Practices	Institution Admins	How to evaluate supplier quotes, verify delivery quality, and rate accurately
Platform Navigation Tour	New Users	Interactive walkthrough of all dashboard features — role-specific
Safety & Compliance	All Users	Electrical safety, data security, and equipment handling standards

19. ORGANIZATION TYPES & LOCATION SYSTEM

Type	Examples	Device ID?	Internal Tech?	Marketplace
Government Institution	MININFRA, REB, RURA, District Offices	Required	Yes	Fallback
Public Hospital	King Faisal, CHUK, District Health Centers	Required	Sometimes	Yes
Public University/School	UR, RP, Secondary Schools	Optional	Rare	Yes
Private Company	MTN Rwanda, BK, Equity Bank, Fasho, etc.	Optional	Rare	Primary
NGO	Local & International NGOs, Foundations	Optional	No	Yes
Religious Institution	Churches, Mosques, Missions	No	No	Yes
Individual / SME	Small offices, shops, repair centers	No	No	Yes
Other	Must self-describe on registration	Optional	Optional	Yes

Rwanda Location Hierarchy — Dynamic Dropdown

All location fields across SMART-2MCE use a dependent dropdown system. Selecting a Province filters Districts; selecting a District filters Sectors, and so on — exactly as used in Irembo.rw and official government forms.

Level	Unit	Details
Level 1	Province	5 provinces: Eastern, Western, Northern, Southern, Kigali City
Level 2	District	30 districts — filtered by selected province
Level 3	Sector	416 sectors — filtered by selected district
Level 4	Cell	2,148 cells — filtered by selected sector
Level 5	Village	14,837 villages — filtered by selected cell

20. DATABASE SCHEMA (PostgreSQL)

Table	Purpose	Key Fields
users	All platform users	id UUID, name, email, role, organization_id, created_at
organizations	All institutions and businesses	id, name, type, province, district, sector, has_internal_tech
devices	Registered devices per org	id, org_id, device_code, name, category, room, condition
materials	Registered materials per org	id, org_id, name, room, quantity, condition
service_requests	Core maintenance requests	id, org_id, device_id, priority, deadline, status, late_flag
marketplace_announcements	Official files for external marketplace	id, request_id, file_url, type (TECH/SUPPLIER/BOTH), status
technician_proposals	External technician job applications	id, request_id, tech_id, proposed_price, status
assignments	Final assignment record	id, request_id, tech_id, supplier_id, type, assigned_at
supplier_quotes	Competitive bids from suppliers	id, request_id, supplier_id, price, delivery_time, status
spare_part_requests	Parts requests within tasks	id, request_id, item_name, quantity, status
messages	Task-scoped communications	id, request_id, sender_id, receiver_id, message, created_at
notifications	All system alerts and reminders	id, user_id, type, message, is_read, created_at
ratings	Post-completion ratings	id, from_user, to_user, request_id, score (1-5), comment
audit_logs	Full immutable action log	id, user_id, action, entity_type, entity_id, timestamp

Status Enumerations

Entity	Status Values
Service Request	CREATED → INTERNAL_REVIEW → OPEN_MARKETPLACE → UNDER_APPLICATION → NEGOTIATION → ASSIGNMENT
Tech Proposal	PENDING → NEGOTIATING → ACCEPTED REJECTED
Supplier Quote	PENDING → ACCEPTED REJECTED
Spare Part Request	OPEN → QUOTED → ORDERED → DELIVERED
Device / Material	WORKING → FAULTY → UNDER_MAINTENANCE → DECOMMISSIONED
Organization	ACTIVE → SUSPENDED PENDING_VERIFICATION



21. TECHNOLOGY STACK & DEVELOPMENT PHASES

Layer	Technology	Why
Frontend	React.js + Material UI + Axios + Redux + React Router	Component-based; professional dashboards; ecosystem with React-Query for data sync
Backend	Node.js + Express.js + REST API	Same JS language as frontend; fast; massive ecosystem; ideal for API services
Database	PostgreSQL	Relational model handles complex joins (requests ↔ technicians ↔ suppliers ↔ ratings)
Authentication	JWT + OTP (2FA) + BCrypt	Stateless sessions; two-factor security; encrypted password storage
Real-Time	Socket.IO	Live countdown updates, instant notifications, negotiation thread updates
File Storage	AWS S3 / Local (MVP)	Marketplace announcement files, certificates, profile photos
Deployment	Render (Backend) + Vercel (Frontend) + Supabase (DB)	Deployment tier available for MVP and demo
Version Control	Git + GitHub	Collaboration, CI/CD pipeline, code review, issue tracking

Development Phases

Phase	Focus	Key Deliverables	Timeline
1	Foundation	Auth, roles, org registration, location dropdowns, JWT	Week 1–2
2	Asset Management	Device registry (ID), material registry (room), CRUD	Week 3
3	Service Requests	Full request lifecycle, time engine, countdown, late detection	Week 4–5
4	Marketplace	Technician marketplace, proposals, negotiation, supplier ratings	Week 6–7
5	Real-Time	Socket.IO notifications, live countdown, dashboard updates	Week 8
6	AI & Analytics	Charts, performance dashboards, AI assistant, query engine	Week 9–10
7	Learning Center	Training modules, guides, role-based content recommendations	Week 11
8	Polish & Deploy	Full testing, mobile optimization, deployment, competitor prep	Week 12



22. INNOVATIONS & SOCIAL IMPACT

Graduate Employment Marketplace	Unemployed graduates find and apply for real, local maintenance jobs. Direct economic impact on youth unemployment.
No Auto-Assignment Rule	Technicians apply and negotiate before starting work. Institution always chooses. Fairness and trust are preserved.
Competitive Supplier Bidding	Suppliers compete transparently through quotes. Institutions select on price, speed, and rating — not relationships.
Smart Time Engine	Every task has a deadline, countdown timer, and multi-level late alert. Delays are detected automatically and escalated.
Rwanda Location Intelligence	Province→Village dropdown hierarchy enables geographic smart matching, proximity filtering, and national analytics.
Marketplace Announcement Files	Official documents required before external access. Adds formality, legal traceability, and institutional accountability.
Multi-Tenant, Role-Adaptive Platform	Public, private, NGO, hospital — each organization type gets an adapted experience within one shared infrastructure.
AI Assistant & Predictive Analytics	Natural-language AI queries, predictive device failure warnings, and smart recommendation engines across all modules.
Integrated Learning Center	Reduces future maintenance volume by educating users on proper device usage, fault detection, and reporting quality.
Full Audit Trail	Every action is logged, timestamped, and attributable. Critical for government transparency and dispute resolution.
Task-Scoped Communication Only	No free chat. All messages are linked to specific tasks — professional, secure, and legally traceable.
National Scale by Design	Multi-tenant architecture, Rwanda location hierarchy, and modular design allow expansion to any African country.



23. COMPETITION PITCH

SMART-2MCE Elevator Pitch:

We built a smart national maintenance ecosystem that simultaneously connects Rwanda’s institutions to skilled graduate technicians and competitive suppliers — through a single, transparent, time-driven digital platform. We solve maintenance delays, graduate unemployment, and overpriced procurement at once — with AI, automation, and accountability built in.

Problem–Solution Matrix

Problem	SMART-2MCE Solution	Measurable Outcome
Institutions lack technicians	Graduate marketplace — on-demand, skill-based	Faster resolution, lower cost
No maintenance accountability	Time engine + audit trail + ratings	Full traceability of every action
Graduate unemployment	Marketplace with proposals, negotiation, Real	Real income for skilled graduates
Overpriced procurement	Competitive supplier bidding	Market-rate pricing, no overcharging
No asset visibility	Device ID + room registry + history	Complete asset lifecycle tracking
Unstructured communication	Task-scoped, role-restricted messaging	No disputes, full records
Manual paper workflows	Digital, automated, service-oriented platform	Data-driven, paperless operations

Scalability & Vision

Dimension	Current	Future Potential
Geographic Scope	Rwanda — all provinces, districts, sectors	East Africa, then continental
Sectors	Public, Private, NGO, Health, Education	Transport, Agriculture, Energy
Actors	3 core types (Inst., Tech., Supplier)	Add Insurance, Training Orgs, Gov. Dashboards
Technology	Web platform (React + Node + PostgreSQL)	Mobile App, QR code scanning, offline mode
Revenue Model	Free MVP — subscription + commission	SaaS, per task organization + premium analytics



24. BUSINESS RULES & LOGOUT FLOW

24.1 Non-Negotiable System Business Rules

Rule 1	INTERNAL PRIORITY	Internal technicians are always offered a task before the external marketplace opens.
Rule 2	NO AUTO-ASSIGNMENT	External technicians must apply manually. The system NEVER automatically assigns them.
Rule 3	NEGOTIATION REQUIRED	Negotiation must occur and institution must formally accept before any external work begins.
Rule 4	DEADLINE IS MANDATORY	Every task must have a deadline. Countdown is always visible to all parties.
Rule 5	LATE FLAG IS AUTOMATIC	System auto-flags LATE when deadline passes without COMPLETED status — no manual override.
Rule 6	ANNOUNCEMENT FILE	Marketplace announcement PDF must be uploaded before a task opens to external technicians.
Rule 7	SUPPLIER COMPETITION	Suppliers compete through quotes. Institution always selects — the system never auto-chooses.
Rule 8	TASK-SCOPED CHAT ONLY	All communication is linked to specific tasks. No free messaging between unrelated parties.
Rule 9	IMMUTABLE AUDIT LOG	Every system action is permanently recorded. Logs cannot be deleted or modified.
Rule 10	MANDATORY RATINGS	After task closure, both institution and technician must submit ratings before task is archived.
Rule 11	TYPE-ADAPTIVE BEHAVIOR	The platform automatically adapts fields, workflows, and displays based on organization type.
Rule 12	DEVICE ID STANDARDS	Public institutions must use structured device codes (e.g., KRH/MH/001). Optional for private.

24.2 Session Management & Logout Flow

Event	System Behavior
User clicks "Sign Out"	JWT token invalidated server-side; session cleared; redirect to landing page
Session timeout (30 min)	Automatic logout with warning at 25 min; unsaved data prompt shown
Browser closed	Session persists if "Remember me" was checked; expires after 7 days
Multiple devices	JWT per device; each can be independently invalidated from account settings
Account suspended	All active sessions immediately terminated; user notified by email
After logout	User sees landing page with option to sign in or register — no data retained in browser



SMART-2MCE — End-to-End Journey:

A visitor lands on the public home page → learns about the ecosystem → registers as Institution, Technician, or Supplier → verifies identity via OTP → accesses their role-specific dashboard → creates or finds work → negotiates and completes tasks → rates and gets rated → builds reputation → grows with the platform → logs out securely. Every step is guided, tracked, accountable, and intelligent.

— End of SMART-2MCE Full Ecosystem Documentation —

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